

A free hand: on scientific institutes and GEM norms

Science must be **unfettered** if it is to be useful

Last week, the government issued a set of orders that scientists have **heralded** as ‘revolutionary’. A major change is in allowing scientific institutions to **bypass** the Government e-Marketplace (GEM), a Commerce Ministry initiative that is meant to **prioritise** made-in-India equipment. GEM norms require all government purchases — from laptops to furniture — to be routed through the GEM-portal, with a **mandate** to buy from the vendor offering the lowest price. While **technocrats** in government **amplified** this bypass as a “landmark” initiative to promote ‘ease of doing research and development,’ the fact is that until GEM-based **procurement** was made mandatory from 2020, the default option was to allow individual scientific institutions the freedom to make their choices regarding the vendors they **procure**. Take for example, sodium chloride. Something as common as table salt must be available in infinite supply and it is only proper that laboratories — they require great quantities for its **myriad** applications in research — source it from the supplier who offers it the cheapest. However, **much like the avatars** of salt — kosher, flat or sea — **are** uncommonly unique to the chef, the **differences** in purity even within common salt **are critical to** scientific research as well as the manufacture of pharmaceuticals. This translates to some vendors being more reliable and, therefore, more preferred.

A major **aspect** of scientific research **is** about being able to reproduce results of experiments described in publications. Often, this requires **fidelity** to the methods and materials of the original **experimenter**. **Given** the challenge of budgets, the **inability** to **source** the right material **results in** experiments being **junked** halfway, or **crimping on** experimental **ambition**, resulting, overall, in a net loss of resources, time and effort. If this is extended to materials more complicated than salt — **precision lathes**, customised lab-produced diamonds, biological molecules, for example — it is easy to understand the **gripe** of scientists. It is understandable, and **pardonable**, when a government experiments with an untested policy and runs into **uncharted waters** or **unknown unknowns**. In the case of GEM, it was a known fact that India lacked an industrial base for sophisticated machinery, and it was **inevitable** that the **hammer-head policy** that saw all products as cookie-cutter nails **would impede** scientific research. India’s scientific ministries are unique in that they are led by scientists, instead of the usual norm of having career **bureaucrats**. This was due to a recognition, **dating back to** the early years of the republic, that while science and technology can be employed to serve the state, science itself is unfettered and must be specially **nurtured** to be useful. A free hand is worth more than two **fettered** arms.

[Practice Exercise]

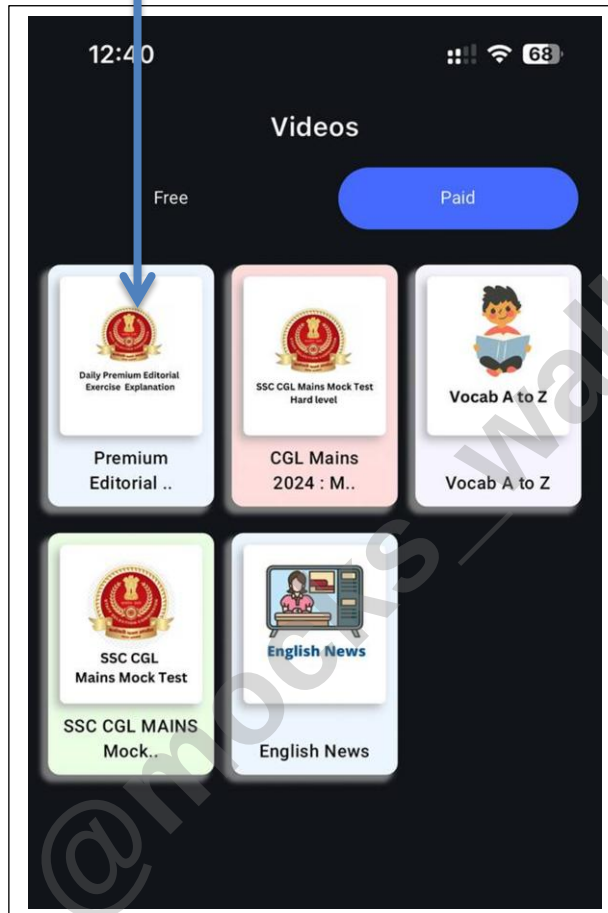
- Red/blue coloring of words in the sentence indicates subject verb relationship; where ‘red’ denotes ‘subject’ and ‘blue’ denotes ‘verb’.

Vocabulary

1. **A free hand** (phrase) – freedom to act completely at one's own discretion. स्वतंत्र निर्णय की छूट
2. **Unfettered** (adjective) – unrestrained, unrestricted, free, uninhibited, unbounded बेपरवाह / बिना रोक-टोक के
3. **Herald** (verb) – announce, signal, proclaim, declare, introduce स्वागत करना
4. **Bypass** (verb) – avoid, skip, circumvent, sidestep, evade दरकिनार करना
5. **Prioritise** (verb) – rank, emphasise, prefer, focus on, highlight प्राथमिकता देना
6. **Mandate** (noun) – order, directive, instruction, requirement, command आदेश / निर्देश
7. **Technocrat** (noun) – expert, specialist, engineer, professional, administrator तकनीकी विशेषज्ञ
8. **Amplify** (verb) – increase, enlarge, magnify, intensify, boost बढ़ा-चढ़ाकर पेश करना
9. **Procurement** (noun) – purchase, acquisition, sourcing, buying, obtaining खरीद
10. **Procure** (verb) – acquire, buy, obtain, secure, source खरीदना
11. **Myriad** (adjective) – numerous, countless, innumerable, endless, many अनेक
12. **Avatar** (noun) – form, incarnation, version, manifestation, embodiment रूप
13. **Critical** (to) (adjective) – crucial, vital, essential, necessary, indispensable अत्यावश्यक
14. **Aspect** (noun) – facet, feature, element, component, angle पहलू
15. **Fidelity** (noun) – accuracy, loyalty, faithfulness, exactness, consistency निष्ठा / सटीकता
16. **Experimenter** (noun) – researcher, tester, scientist, analyst, investigator प्रयोगकर्ता
17. **Given** (preposition) – considering, assuming, in view of, due to, taking into account को देखते हुए
18. **Source** (verb) – obtain, procure, acquire, get, secure प्राप्त करना
19. **Result in** (phrasal verb) – lead to, cause, bring about, produce, generate का परिणाम होना
20. **Junk** (verb) – discard, throw away, scrap, dump, dispose of बेकार करना
21. **Crimp on** (phrasal verb) – limit, restrict, reduce, hinder, constrain रोक लगाना

22. **Ambition** (noun) – aspiration, aim, goal, objective, desire महत्वाकांक्षा
23. **Precision lathes** (noun) – high-accuracy cutting machines, engineering tools, CNC lathes, fine lathes, machine tools उच्च सटीकता वाली खराद मशीनें
24. **Gripe** (noun) – complaint, objection, protest, grievance, criticism शिकायत
25. **Pardonable** (adjective) – forgivable, excusable, acceptable, justifiable, understandable क्षम्य / माफ़ किया जा सकने वाला
26. **Uncharted waters** (noun) – a situation that is new, unfamiliar, and potentially risky or difficult to navigate अनजाना क्षेत्र
27. **Unknown unknowns** (phrase) – things that are not known, and more importantly, not even known to be unknown अज्ञात अनिश्चितताएँ
28. **Inevitable** (adjective) – unavoidable, certain, inescapable, destined, bound to happen अनिवार्य / अपरिहार्य
29. **Hammer-head policy** (noun) – rigid approach, inflexible strategy, one-size-fits-all method, blunt policy कठोर और एकसमान नीति
30. **Impede** (verb) – hinder, obstruct, block, hamper, delay बाधा डालना
31. **Bureaucrat** (noun) – official, administrator, civil servant, government officer, policymaker नौकरशाह
32. **Date back to** (phrase) – originate in, go back to, trace to, stem from, come from की शुरुआत होना
33. **Nurture** (verb) – foster, cultivate, support, encourage, nourish बढ़ावा देना
34. **Fettered** (adjective) – restricted, bound, chained, limited, restrained जकड़ा हुआ / बाधित

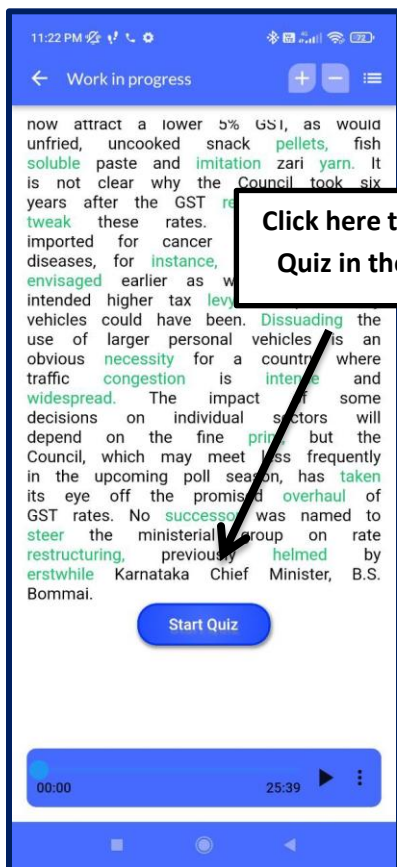
Video solution of Practice Exercise from this Editorial is exclusively available on App.



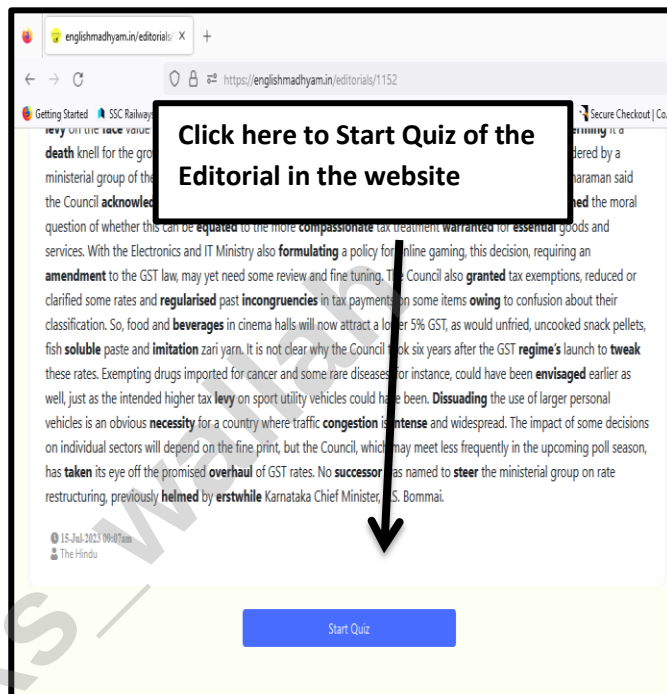
Summary of the Editorial

1. The Indian government recently allowed scientific institutions to bypass the mandatory Government e-Marketplace (GEM) for procurement.
2. Scientists welcomed this move as a “revolutionary” and positive step for research and development.
3. GEM was introduced to ensure government purchases favored Indian-made products at the lowest price.
4. Before GEM became mandatory in 2020, institutions had autonomy to choose their vendors.
5. The GEM system, while efficient for general items, fails to consider the nuanced requirements of scientific materials.
6. For example, even simple chemicals like sodium chloride vary in purity, which is critical for scientific research.
7. The lowest price is not always the best criterion when quality and consistency are vital in research.
8. Scientific experiments often depend on exact replication of materials and methods to ensure reproducibility.
9. Constraints on sourcing the right materials lead to failed experiments or limited research scope.
10. In complex scientific equipment — like precision tools, custom materials, or bio-molecules — GEM's uniform policy proves highly restrictive.
11. The GEM policy didn't account for India's limited manufacturing base for high-end scientific tools.
12. Applying a one-size-fits-all approach through GEM hampered the freedom and efficiency of research institutions.
13. The editorial acknowledges that policy experimentation is normal, but known shortcomings in infrastructure should have been considered.
14. India's scientific ministries are uniquely led by scientists, unlike most bureaucratic structures.
15. The editorial emphasizes that science thrives only when it is free and unburdened by rigid, ill-suited policies.

Practice Exercise: SSC Pattern Based



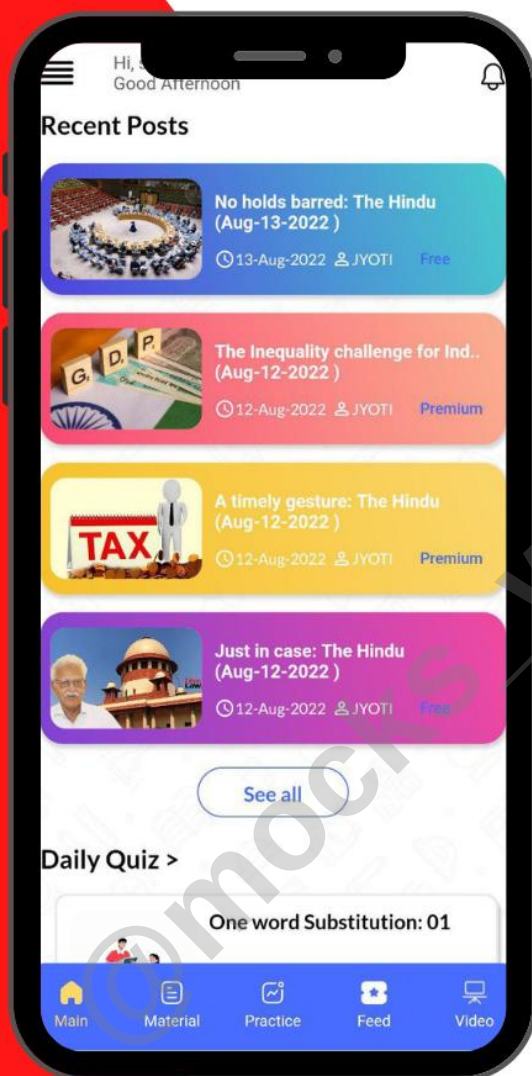
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